

Subhraneel Goswami

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EDUCATION

Institute of Engineering and Management

Bachelor of Computer Applications

Kolkata, IN

Jun. 2023 – May 2026

EXPERIENCE

Jobsforce.ai

Full Stack Developer Intern

Mar 2025 – Sep 2025

SF, (Remote)

- Developed a full-stack AI-powered Chrome extension using Plasmio.
- Implemented real-time communication between LLMs and users using WebSockets and Node.js.
- Built scalable frontend data handling for 5,000+ job listings using pagination and infinite scroll.
- Designed and implemented secure authentication APIs with role-based access control middleware.
- Worked across frontend and backend systems to ship production-ready features.

Kasukabe Labs (Web and App Dev Agency)

Fullstack/Applied AI Intern

Oct 2025 – Jan 2026

Kolkata, India, (Remote)

- Contributed to a production AI video generation pipeline built with Python and FastAPI, architected as a collection of independent async microservices for dialogue generation, character design, audio synthesis, and video assembly.
- Built and maintained dialogue audio generation services integrating LLMs and ElevenLabs TTS, handling high-volume API interactions with resilient error handling and asynchronous processing.
- Developed a character concept art generation service using Gemini image generation and Banana API, executing concurrent generation workloads using `asyncio.gather` to improve throughput and reduce latency.
- Implemented background audio generation workflows and service-to-service communication patterns across the video generation pipeline.
- Worked extensively with asynchronous Python, external AI providers, batching strategies, retry handling, and production API integrations.
- Refactored monolithic services into modular, maintainable components and authored technical documentation covering architecture, workflows, and service contracts.

PROJECTS

AI Terminal Coding Agent | *TypeScript, OpenRouter, Gemini, Tool Calling, Node.js*

- Built a CLI-based AI coding agent capable of autonomous tool usage for navigating and modifying codebases.
- Implemented tool-calling architecture enabling the agent to dynamically invoke filesystem and bash tools.
- Integrated Git tools enabling the agent to reason over version history, staged changes, push/pull, create issues and PRs.
- Built persistent agent memory that grows with user and project preferences across sessions.
- Designed core agent loop supporting streamed responses, multi-step reasoning, and token usage tracking.
- Implemented filesystem tools (`read_file`, `search_files`, `write_file`, `edit_file`) with typed I/O and safe failure handling.
- Added human-in-the-loop diff approval system before applying file edits to prevent destructive actions.
- Implemented path traversal protection ensuring the agent cannot operate outside the project root.
- GitHub: github.com/subhraneel2005/sidequests

Usecerebr | *Python, FastAPI, Qdrant, Gemini 2.5 Flash, Redis, Docker*

- Built a personal knowledge graph and second brain that semantically connects ideas across sources — YouTube videos, research papers, blogs, articles, and X posts.
- Implemented a multilingual RAG pipeline ingesting content (English + Hindi) into a vector database for cross-source semantic QA.

- Chunked and embedded content using all-MiniLM-L6-v2 via Sentence Transformers; stored vectors in Qdrant with cosine similarity retrieval.
- Built a FastAPI backend with query endpoints that embed user questions, retrieve top-K chunks, and generate grounded answers using Gemini 2.5 Flash.
- Used Redis as a job queue for background ingestion pipelines, enabling async processing at scale.
- Dockerized the full stack with Docker Compose for one-command self-hosting.
- GitHub: github.com/subhraneel2005/usecerebr

Agentic AI Study Platform | *Next.js, TypeScript RSC, Server Actions, Vercel AI SDK, Gemini*

- Built an open-source, student-first productivity platform to help students manage real academic workflows.
- Implemented credit-based AI usage with 300 free credits and intelligent token costing based on tool complexity.
- Designed a fallback system allowing users to plug in their own Gemini API key once credits are exhausted.
- Used React Server Components and Server Actions for secure, optimized data mutations and rendering.
- Live: study-toolkit.vercel.app GitHub: github.com/subhraneel2005/study-toolkit

Multi-Agent Customer Support System | *Python, LangGraph, FastAPI, Qdrant, Pydantic, Redis, Docker*

- Built a production-grade multi-agent customer support system using LangGraph with triage-based routing across billing, technical support, and human escalation agents.
- Implemented a triage agent performing intent classification, entity extraction, and sentiment analysis to route queries to the correct specialist via conditional graph edges.
- Built a RAG-powered technical support agent using Qdrant vector search (cosine similarity, top-K retrieval) with grounded, citation-backed responses and confidence scoring.
- Developed a billing agent with invoice verification tooling, KB-grounded policy lookup, and structured escalation logic using Pydantic output models.
- Designed a handover protocol with serializable context payloads, audit logging, and clean agent-to-agent state transfer across the graph.
- Built a FastAPI REST surface (POST `/conversations`, POST `/conversations/{id}/messages`, GET `/conversations/{id}`) with YAML-driven guardrails and structured JSON logging with trace IDs.
- GitHub: github.com/subhraneel2005/multi_agent_cs_system

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Go (learning), Java, SQL

Frontend: React, Next.js (App Router, RSC), Expo, HTML5, CSS3, Tailwind CSS, shadcn/ui

Backend: Node.js, FastAPI, Bun, REST APIs, WebSockets, AsyncIO, Microservices, Server Actions, Authentication, Authorization, JWT, Redis

Databases & ORM: PostgreSQL, MongoDB, Prisma

AI & LLM Systems: OpenAI API, Gemini API, Vercel AI SDK, LangGraph, Tool Calling, Streaming, AI Agents, RAG, Vector Databases, Qdrant, Sentence Transformers

Infrastructure: Linux, Docker, Kubernetes (learning), Vercel, AWS, GCP, NeonDB

Developer Tools: Git, GitHub, Cursor, Claude Code, Opencode